



Expanded Kuro Siwo

Data Catalog for Flooded Environment Analysis

GROUP

正在輸入中...

MEMBERS

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Content

1. Project Objective
2. Workflow Overview
3. Dataset Processing
 1. Dataset Brief Introduction
 2. Environment Setup
 3. Data Engineering – Format & partitioning strategy
 4. COG Tiling Strategy Comparison
 5. COG and Zarr Comparison (Sentinel-2)
 6. STAC Structure & Catalog & Metadata & Extensions
4. Downstream Application
5. Benefits & Potential for Further Expansion



Project Overview

Background and Objective

Background

Kuro Siwo: A global multi-temporal SAR dataset for rapid flood mapping

Number of Events	43
Total Area	33 billion square meters
Included Data	SAR SLC & GRD, labeled data of floods / permanent water / no water, DEM

✓ High-quality expert-annotated data.



✓ A vast amount of globally diverse SAR data

Objective:

以 Kuro Siwo 資料集為基礎，加入同樣與淹水有關之地理資料，並將原始資料轉化為**高效、可搜索且具備機器可讀性**的擴充資料集，以支援下游的淹水環境分析與淹水辨識模型

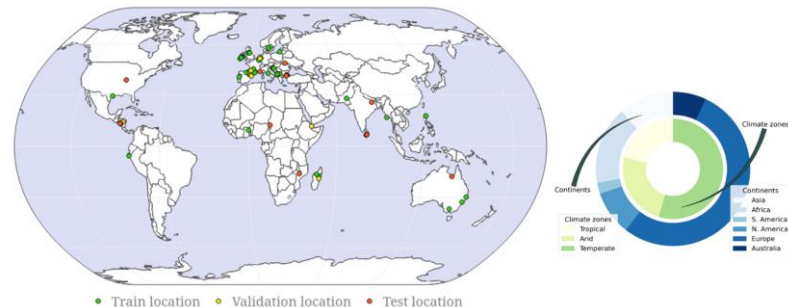


Figure 1: Spatial distribution of Kuro Siwo events. The test dataset encompasses flood events from entirely unseen locations on Earth.

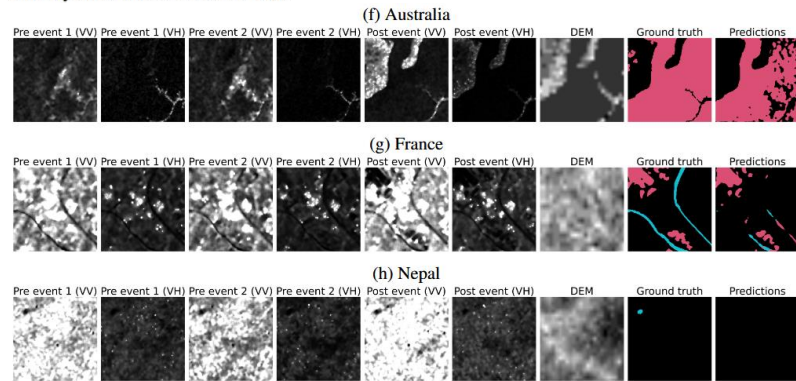
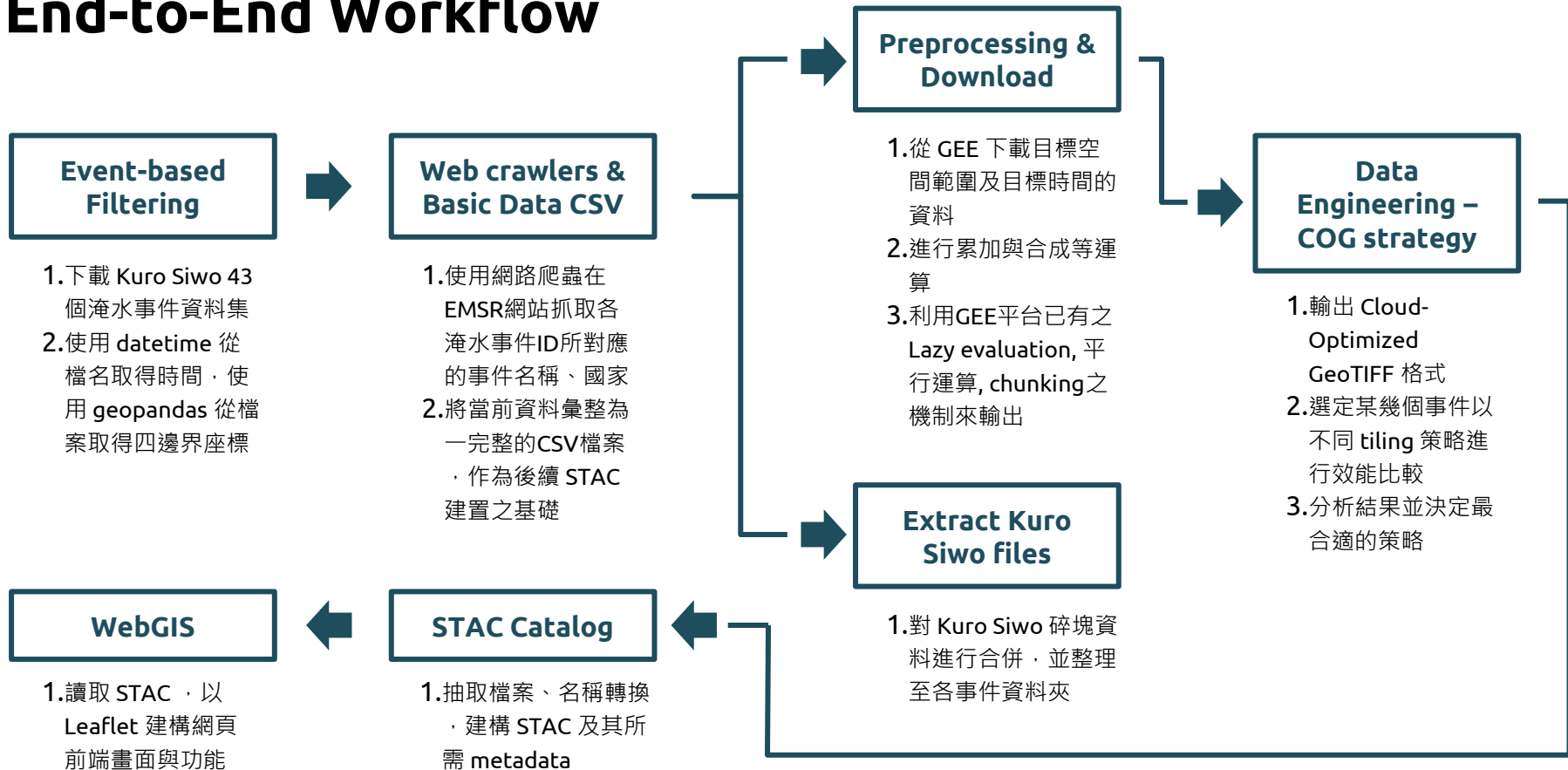


Figure 6: Kuro Siwo GRD samples. Flood is marked in **purple**, permanent waters in **cyan** and non-water areas in **black**. The predictions come from the best performing model in BlackBench, i.e UNet-ResNet50.

End-to-End Workflow





Dataset Brief Introduction

Dataset Brief Introduction

The data is intended for Flood Detection and Monitoring!

	Sentinel-1 SAR GRD	Sentinel-2 MSI L2A	IMERG	Dynamic World V1	SRTM	Flood event labeled data	Manual labeled data
Characteristic	Synthetic Aperture Radar	Multi-spectral imaging for land monitoring	Global Precipitation Measurement	Near real-time global land cover / land use classification	Digital Elevation Models (DEMs)	Labeled data	Labeled data
Spatial Resolution	10 m	10 m	11132 m	10 m	10 m	(.shp)	10 m
Time Resolution	6 Days	5 Days	30 Minutes	5 Days	--	--	--
Selected Variables	VV, VH	B4 (Red), B3 (Green), B2 (Blue), B8 (NIR)	precipitation	water, trees, grass, flooded_vegetation, crops, shrub_and_scrub, built, bare, snow_and_ice	elevation	flooded area	0 (permanent water) 1 (no water) 2 (water) 3 (flood)
Source	Kuro Siwo dataset	GEE	GEE	GEE	Kuro Siwo dataset	Kuro Siwo dataset	Kuro Siwo dataset
Preprocess work we did	Merge the files to one file	Mosaic stitching images from three days prior and later	Sum the results of the previous 7 days	Year-scale composite time window; Generate both probabilistic and categorical products	Merge the files to one file	--	Merge the files to one file



Environment Setup

Environment Setup

Hardware Environment

CPU

AMD Ryzen 9
5950X 16-Core
Processor (Threads
per core: 2).

RAM

78 GB

Disk

NAS storage - 84 TB

OS

Linux (Ubuntu)

Parallel Computing

引入 Dask 框架，配置 4 個 workers，每個 worker 配置 8 緒，以進行大規模影像轉換的平行處理與效能基準測試



Dependency Management

```
chunen [SSH: 3090]
└─ course_BigData
  └─ .venv
  └─ CIE5158_W1_hands-
  └─ Final
  └─ stac-browser
  └─ W2
  └─ W3
  └─ W4
  └─ .gitignore
  └─ .python-version
  └─ main.py
  └─ pyproject.toml
  └─ README.md
  └─ uv.lock

course_BigData > pyproject.toml
1  [[project]]
2  name = "w1-gee-hands-on"
3  version = "0.1.0"
4  description = "Add your description here"
5  readme = "README.md"
6  requires-python = ">=3.11"
7  dependencies = [
8      "cartopy>=0.25.0",
9      "cf-xarray>=0.10.11",
10     "cftime>=1.6.5",
11     "cmocean[plots]>=4.0.3",
12     "dask>=2026.1.2",
13     "earthengine-api>=1.7.15",
14     "geemap>=0.37.2",
15     "geopandas>=1.1.3",
16     "geoviews>=1.15.1",
17     "h5netcdf>=1.8.1",
18     "hvplot>=0.12.2",
19     "jupyter-bokeh>=4.0.5",
20     "matplotlib>=3.10.8",
21     "netcdf4>=1.7.4",
22     "pystac-client>=0.9.0",
23     "rasterio>=1.4.4",
24     "rich>=14.3.3",
25     "rio-cogeo>=7.0.2",
26     "rioxarray>=0.19.0",
27     "scipy>=1.17.1",
28     "shapely>=2.1.2",
29     "stac-validator>=4.1.0",
30     "stackstac>=0.5.1",
31     "xarray>=2026.2.0",
32     "xee>=0.0.24",
33     "zarr>=3.1.5",
34 ]
35
```



Dataset Processing

Data Engineering – Format & partitioning strategy

Unit File Design



Based on Flood Event

- 基於 KuroSiwo 資料集結構
- 便於查找篩選

Selected Format



COG

Cloud Optimized GeoTIFF

- 基於離散事件資料
- 適合 WebGIS 展示

Tiling Strategy Design

Downstream task: WebGIS

其特性為：

1. 小範圍 (tile-based) 讀取
2. 隨機存取 (random access)
3. 多使用者併發

影響因子：

1. Tiling
2. Compression
3. Overview



- **Strategy A : Orig**
 - 取消 CloudOptimize 之標準 GeoTIFF 檔案
- **Strategy B : Default**
 - 使用 GEE default COG 輸出之檔案
- **Strategy C : 256**
 - 使用 256 x 256 tiles 之 COG 檔案
- **Strategy D : 512**
 - 使用 512 x 512 tiles 之 COG 檔案
- 針對 Strategy C & D，固定 Compression 方式為 "DEFLATE"，Overview 重採樣方式為 "average"

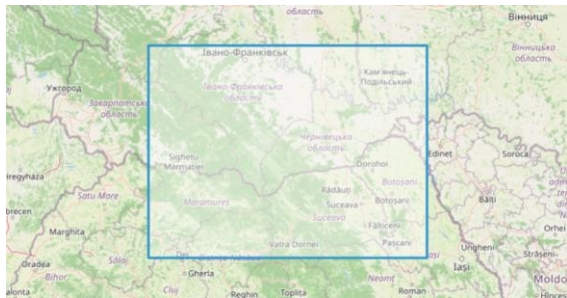
Data Engineering – Format & partitioning strategy

Key points of comparison

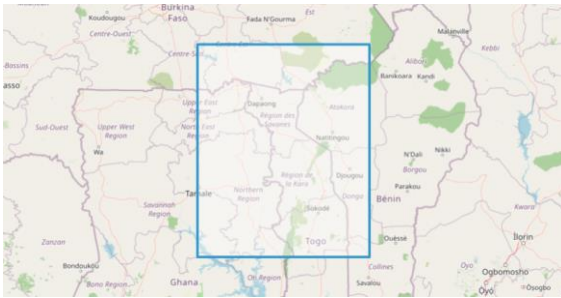
1. 檔案容量
2. 讀取及運算所需時間
3. CPU 占用程度
4. 記憶體占用程度

- 使用 **Random** 隨機讀取窗格，並模擬讀取**100個窗格**來測試，窗格大小為256x256 (或原圖片大小)
- 上述運算動作**重複 5 次**，並選擇中位數的數值作為測試結果
- 使用 **dask** 建立任務清單，再以平行運算的方式派發到 **4 個 workers**、每個workers各有 **8 個 threads**
- 挑選三場發生於不同類型地區的淹水事件來測試

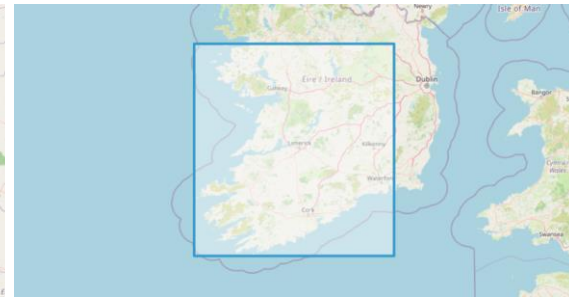
EMSR445_Flood_in_Romania



EMSR470_Flood_in_Togo

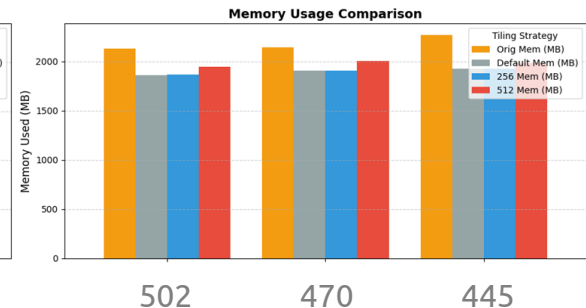
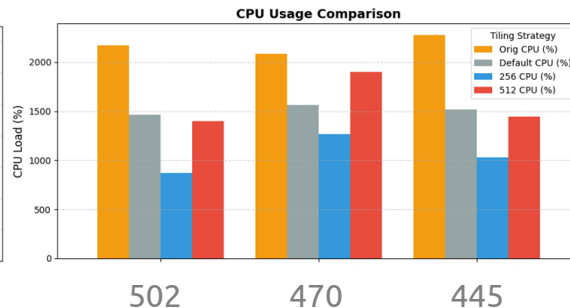
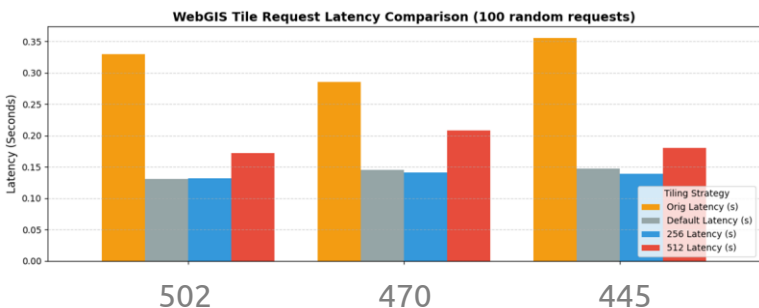


EMSR502_Flood_in_Southern_Ireland



COG Tiling Strategy Comparison (Sentinel-2)

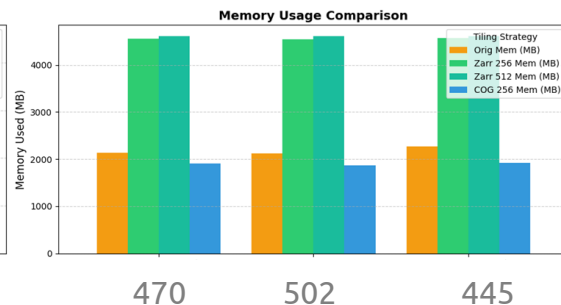
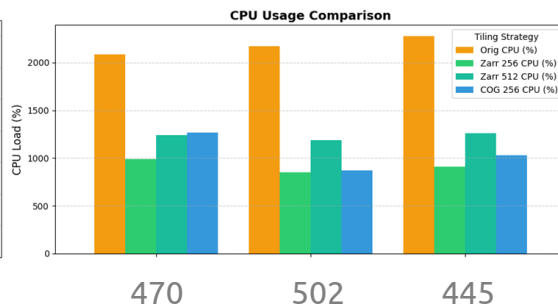
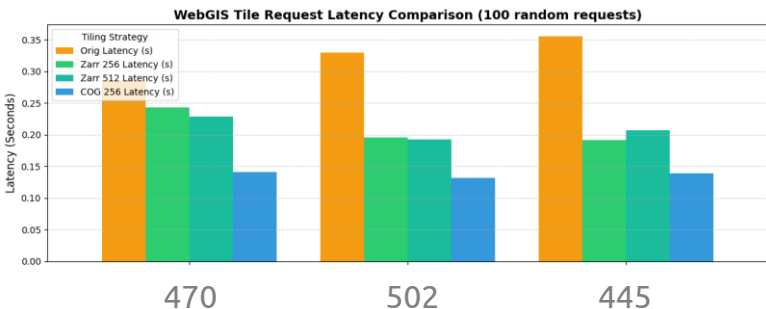
ID	File Size (MB)				Latency (s)				CPU Usage (%)				Memory Usage (MB)			
	Orig	Default	256	512	Orig	Default	256	512	Orig	Default	256	512	Orig	Default	256	512
502	324.96	245.67	158.36	160.38	0.330	0.131	0.132	0.172	2173.2	1466.0	870.4	1397.0	2126.9	1861.0	1864.6	1941.4
470	1098.32	1306.20	1069.24	1082.72	0.285	0.145	0.141	0.208	2086.6	1563.2	1265.2	1899.1	2142.3	1905.5	1904.6	2001.6
445	553.48	616.74	356.60	358.44	0.356	0.147	0.139	0.180	2279.6	1520.5	1029.5	1443.6	2269.3	1923.6	1921.9	1983.0
AVG	658.92	722.87	528.07	533.85	0.324	0.141	0.137	0.187	2179.8	1516.6	1055.0	1579.9	2179.5	1896.7	1897.0	1975.3



- Default, 256 x 256 tiles, 512 x 512 tiles 相較於一般 GeoTIFF 在各指標有明顯提升，且在這種大容量檔案更為明顯
- 256 x 256 tiles 在 File Size ,Latency 及 CPU Usage 有相當明顯的優勢，不僅更節省空間，也能避免不必要的運算負擔
- 本專案欲選定 **256 策略**作為 Sentinel-2 格式選擇，缺點是在較大地區範圍會造成單位面積 IO 次數增加，但以本專案淹水事件而言，此策略在多數地區皆能達降低後端伺服器的運算壓力與儲存成本之效果

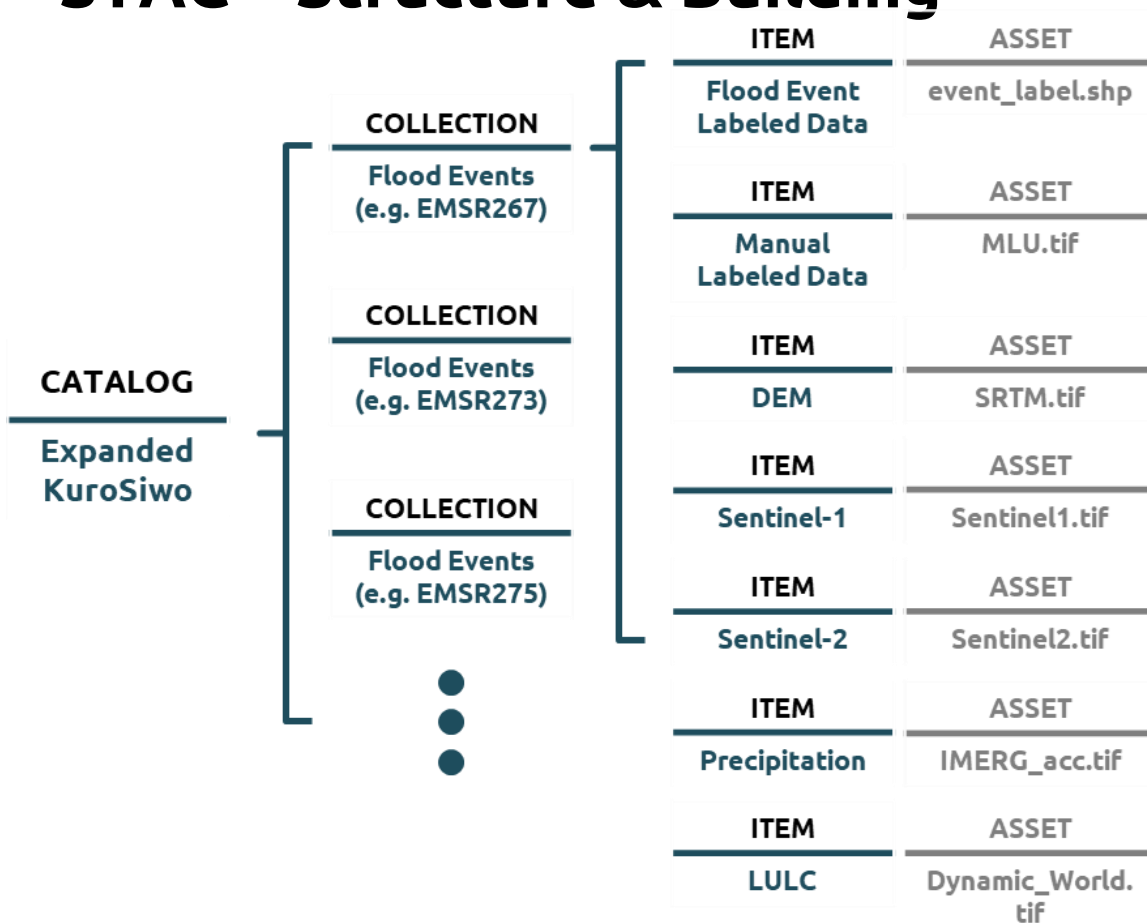
COG and Zarr Comparison (Sentinel-2)

ID	File Size (MB)				Latency (s)				CPU Usage (%)				Memory Usage (MB)			
	Orig	256 Zarr	512 Zarr	256 COG	Orig	256 Zarr	512 Zarr	256 COG	Orig	256 Zarr	512 Zarr	256 COG	Orig	256 Zarr	512 Zarr	256 COG
470	1072.58	684.33	683.07	1044.18	0.285	0.243	0.229	0.141	2086.6	987.5	1240.9	1265.2	2142.3	4551.5	4615.8	1904.6
502	317.34	74.68	74.25	154.65	0.330	0.196	0.193	0.132	2173.2	847.4	1185.0	870.4	2126.9	4549.0	4612.6	1864.6
445	540.51	202.13	200.74	348.25	0.356	0.192	0.207	0.139	2279.6	907.6	1261.9	1029.5	2269.3	4565.3	4603.9	1921.9
AVG	643.48	320.38	319.35	515.69	0.324	0.210	0.210	0.137	2179.8	914.2	1229.3	1055.0	2179.5	4555.3	4610.8	1897.0



- Zarr 相較於一般 GeoTIFF 在前三指標有明顯提升，其中，File Size 和 CPU Usage 上更勝過 COG，但 CPU 方面差距不大
- 在 Latency 方面仍是 COG 較具優勢，在 Memory Usage 方面 COG 更是相較於 Zarr 有兩倍以上優勢
- 本專案最終仍選定 **256 COG** 策略作為 Sentinel-2 格式選擇，雖然在硬碟儲存與原始傳輸體積上輸給 Zarr，但考量系統需要服務多位使用者並維持極速的地圖響應，此策略的「低延遲」與「穩定的記憶體佔用」較為關鍵

STAC – Structure & Building



Key Building Techniques

■ Fuzzy File compilation

以 Copernicus 網站上 EMSR 淹水事件編號及其附帶資訊來建構 Collections，再解析各檔案檔名以彙整各檔案至所屬淹水事件

■ Dynamic Metadata Extraction

利用 rioxarray 與 geopandas 對檔案自動萃取其 EPSG、Shape、Transform

■ Adding Keyword

使用 geopandas 設定各事件洲別，並新增於 STAC 上，讓使用者快速篩選

Browse

Expanded-KuroSiwo

Expanded-KuroSiwo

Description

A comprehensive, STAC-compliant, multi-temporal dataset curated for global rapid flood mapping. This catalog expands upon the original Kuro Siwo sources around individual flood events (Collections).

Available Asset Types

Each event collection integrates multiple distinct items aligned precisely to the event's geographical and temporal footprint:

Sentinel-1 (S1): Synthetic Aperture Radar (SAR) imagery with dual-polarization (VV/VH).

Sentinel-2 (S2): High-resolution multispectral optical imagery with B2, B3, B4, B8 bands.

Precipitation: Accumulated rainfall metrics retrieved from NASA GPM IMERG Final Run L3.

LULC (Dynamic World): Near real-time 10m Land Use / Land Cover label map and 9 classes probability map.

DEM (SRTM): Downscaling using Shuttle Radar Topography Mission 1 arc second Global data.

MLU: Manual labeled updated data of flood, water body, permanent water body and no water area of Kuro Siwo dataset.

Flood Event Labels: Delineated ground truth annotations of flooded surface areas of Kuro Siwo dataset.

Specifications

This dataset ensures robust standardization by incorporating multiple STAC Extensions:

Projection: Real-time EPSG, Shape, and Transform metrics extracted dynamically.

Electro-Optical (EO): Band mappings for optical imagery.

Classification: Standardized Dynamic World terrestrial properties mapping.

Scientific: Citations and DOI links to primary research literature.

Catalogs 43

TilesListAscendingDescending

Filter catalogs by title, description or keywords

Select keywords

EMSR118_Flood_in_Spain

Flood event EMSR118_Flood_in_Spain

2015-02-03 0:00:00 UTC

EMSR130_Flood_in_Myanmar

Flood event EMSR130_Flood_in_Myanmar

2015-09-13 0:00:00 UTC

EMSR147_Flood_in_Cumbria

Flood event EMSR147_Flood_in_Cumbria

2015-12-10 0:00:00 UTC

EMSR205_Floods_in_Sabaragamuwa_West_Sri_Lanka

Flood event EMSR205_Floods_in_Sabaragamuwa_West_Sri_Lanka

2017-05-30 0:00:00 UTC

EMSR267_Flood_in_Silute_Lithuania

Flood event EMSR267_Flood_in_Silute_Lithuania

2/2/2018, 12:00:00 AM UTC

EMSR273_Flood_in_North_Western_Albania

Flood event EMSR273_Flood_in_North_Western_Albania

2018-03-11 0:00:00 UTC

EMSR277_Flood_in_Thrace_Greece

EMSR279_Flood_in_the_Ebro_river_basin_Spain

EMSR321_Flood_in_Honduras_and_Nicaragua

Expanded-KuroSiwo

Language: English

EMSR267_Flood_in_Silute_Lithuania

UpSourceShare

Description

Flood event EMSR267_Flood_in_Silute_Lithuania

Europe

Licenseother

Temporal Extent2018-02-02 0:00:00 UTC

Items 11AscendingDescending

flood_event_label

2018-02-02 0:00:00 UTC

LULC

2018-02-02 0:00:00 UTC

MLU

2018-02-02 0:00:00 UTC

Precipitation

2018-02-02 0:00:00 UTC

Sentinel-1

2018-02-02 0:00:00 UTC

Sentinel-2

2018-02-02 0:00:00 UTC

SRTM

2018-02-02 0:00:00 UTC

Map

Powered by STAC Browser 5.0.0-beta.1

Expanded-KuroSiwo

Language: English

Sentinel-2

CollectionUpSourceShare

Collection

EMSR267_Flood_in_Silute_Lithuania

Flood event EMSR267_Flood_in_Silute_Lithuania

2018-02-02 0:00:00 UTC

Metadata

General

Time of Data begins2018-02-02 0:00:00 UTC

Time of Data ends2018-02-02 0:00:00 UTC

CategorySentinel-2

Time of Data2018-02-02 0:00:00 UTC

Electro-Optical

Name	Common Name	Description
B4	red	Red
B3	green	Green
B2	blue	Blue

Assets

> s2_267_201707271095429

> s2_267_201802021095432

> s2_267_201708291100026

Additional Resources

Citation information

DOI 10.5270/S2_26n7jr4

DOI 10.1016/j.rse.2018.04.031

STAC – Extension & Metadata

Extension used to describe data

1. ScientificExtension
2. ProjectionExtension
3. EOExtension
4. ClassificationExtension

General

Time of Data begins	2018-02-02 0:00:00 UTC
Time of Data ends	2018-02-02 0:00:00 UTC
Category	Precipitation
Time of Data	2018-02-02 0:00:00 UTC

Electro-Optical

Spectral Bands		
Name	Common Name	Description
precipitation	precipitation	Accumulated Precipitation

Projection

Code	EPSG:4326
Image Dimensions	19 × 37
Transformation Matrix	[0.1000004574; 0, 19.9000910282] [0; -0.1000004574; 56.2002570746]

Scientific

DOI	10.5067/GPM/IMERG/3B-H4V07
Citation	Huffman, G.J., et al. (2023). GPM IMERG Final Run L3 Hourly 0.1 degree x 0.1 degree V07. NASA GES DISC. Data (2014-2022) retrieved from Google Earth Engine (NASA/GPM_L3/IMERG_V07).

Precipitation

General

Time of Data begins	2018-02-02 0:00:00 UTC
Time of Data ends	2018-02-02 0:00:00 UTC
Category	S2
Time of Data	2018-02-02 0:00:00 UTC

Electro-Optical

Spectral Bands		
Name	Common Name	Description
B4	red	Red
B3	green	Green
B2	blue	Blue
B8	nir	NIR

Projection

Code	EPSG:3857
Image Dimensions	11239 × 23296
Transformation Matrix	[10; 0; 2218040] [0; -10; 7352900]

Scientific

DOI	10.5270/S2_-26n7jr4
Citation	Copernicus Sentinel data (2015-2022) processed by ESA. Retrieved from Google Earth Engine (COPERNICUS/S2_SR_HARMONIZED).
Publications	Publication: Claverie, M., et al. (2018). The Harmonized Landsat and Sentinel-2 (HLS) product: Algorithms and methods. Remote Sensing of Environment. DOI: 10.1016/j.rse.2018.04.031

S2

General

Time of Data begins	2018-02-02 0:00:00 UTC
Time of Data ends	2018-02-02 0:00:00 UTC
Category	LULC
Time of Data	2018-02-02 0:00:00 UTC

Classification

Classes		
Color	Value	Identifier
#4169E1	0	water
#9932CC	1	trees
#8B8B33	2	grass
#7788CC	3	flooded_vegetation
#FF6600	4	crops
#FFCC00	5	shrub_and_scrub
#CC0000	6	built
#FF9900	7	bare

Electro-Optical

Spectral Bands		
Name	Common Name	Description
LULC	class	Dynamic World Land Use / Land Cover class

Projection

Code	EPSG:4326
Image Dimensions	731 × 1501
Transformation Matrix	[0.0023996569; 0; 19.9250372436] [0; -0.0024003068; 56.1244744043]

Scientific

DOI	10.1038/41597-022-01307-4
Citation	Brown, C.F., et al. (2022). Dynamic World. Near real-time global 10 m land use land cover mapping. Scientific Data. Data (2021-2023) retrieved from Google Earth Engine.

LULC

General

Time of Data begins	2018-02-02 0:00:00 UTC
Time of Data ends	2018-02-02 0:00:00 UTC
Category	MLU
Time of Data	2018-02-02 0:00:00 UTC

Classification

Classes		
Color	Value	Identifier
#00FFFF	0	permanent water
#000000	1	no water
#00FF00	2	water
#FF00FF	3	flood

Electro-Optical

Spectral Bands		
Name	Common Name	Description
label	class	Manual Labeling Update

Projection

Code	EPSG:3857
Image Dimensions	4928 × 5824
Transformation Matrix	[10; 0; 2358095] [0; -10; 7445345]

Scientific

DOI	10.52202/079017-1204
Citation	@inproceedings{NEURIPS2024_43612b06,author = {Bountos, Nikolaos Ioannis and Sdraka, Maria and Zavras, Angelos and Karavias, Andreas and Karasante, Ilektra and Herekakis, Themistocles and Thanasou, Angeliki and Michail, Dimitrios and Papoutsis, Ioannis},booktitle = {Advances in Neural Information Processing Systems},editor = {A. Globerson and L. Mackey and D. Belgrave and A. Fan and U. Paquet and J. Tomczak and C. Zhang},pages = {38105-38121},publisher = {Curran Associates, Inc.},title = {Kuro Siwo: 33 billion m^2 under the water. A global multi-temporal satellite dataset for rapid flood mapping},url = {https://proceedings.neurips.cc/paper_files/paper/2024/file/43612b0662cb64986ed859fd6ebafe-Paper-Datasets_and_Benchmarks_Track.pdf},volume = {37},year = {2024}}

MLU



Downstream Application

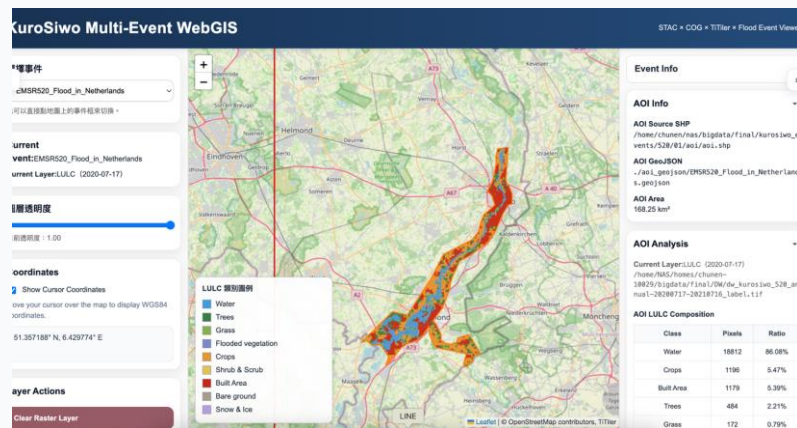
Task Selection: WebGIS Platform

Platform Objectives

- Interactive Visualization:**
Build an engaging geospatial data display system.
- Unified Interface:**
Browse multiple flood events within a single, streamlined UI.
- Comprehensive Insights:**
Access spatial extent, temporal data, and relevant layers instantly.

Core Advantage

WebGIS goes beyond simple data analysis, offering direct data integration and impactful visual results.



Role of STAC in the WebGIS Workflow

Data Integration & Loading

STAC Indexing:

作為多來源淹水事件資料的標準化索引架構

Metadata Loading:

以前端讀取 catalog.json 的方式載入事件 metadata

Path Mapping:

透過 hrefToLocalPath() 映射 STAC 資產與實際檔案路徑

Real-time Tiling:

TiTiler 即時讀取 raster 並切成 map tiles

Front-end Interaction (Leaflet)

使用者可在地圖上：

- 選擇不同淹水事件並縮放至範圍
- 切換降雨、土地利用、Sentinel-2 圖層
- 查看圖例與中繼資料 (Metadata)
- 使用工具欄進行 AOI 統計分析



右側資訊欄降雨統計資料概況



Benefits & Potential for Further Expansion

Benefits & Potential for Further Expansion

大規模處理效率

原始資料：
高解析度、高容量，運算上面臨較大的IO負擔與延遲

本專案改進：
引入 **Dask**、**256 x 256 tiling COG**，使下游應用（如 WebGIS）僅需抓取特定小塊Tile資料，無需下載 GB 級的大檔，提升效率

資料維度擴充

原始資料：
僅有 SAR 與 labeled data

本專案改進：
整合了光學影像、降雨與土地覆蓋，「一站式」地讓研究者能更容易探討多維資料對淹水範圍、深度的具體影響

標準化&可攜性

原始資料：
檔案分散於各處且格式不一

本專案改進：
利用 **STAC catalog** 與相對路徑技術，實現了標準化的索引架構，能直接提供給其他研究團隊使用

Potential for Further Expansion:

- 可擴展至近即時監測，將 GEE 自動抓取的最新影像透過此流程即時推送
- 不限於 Kuro Siwo，自動化導入全球更多淹水事件，打造全球最大的 STAC 兼容淹水環境資料庫